DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING
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Experiment No.: 8

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Semester: 5TH
Date of Performance:
Subject Name: Competitive Coding – II
Subject Code: 24CSP-305

Experiment No: 8.1

Aim: To write a C++ program to calculate the number of distinct ways to climb a staircase.

Objective: To understand dynamic programming and solve a problem using an iterative approach.

Problem Statement: You are climbing a staircase with n steps. You can climb either 1 or 2 steps at a time. Find the number of distinct ways to reach the top.

LeetCode Link: https://leetcode.com/problems/climbing-stairs/submissions/2182140990/?utm_source=chatgpt.com

Procedure/Algorithm/Pseudocode:

1. Start.
2. If n is less than or equal to 2, return n.
3. Initialize a = 1 and b = 2.
4. Iterate from step 3 to n.
5. Calculate c = a + b.
6. Update a = b and b = c.
7. Return b.
8. Stop.

Code:

```
class Solution {

public:

   int climbStairs(int n) {

      if (n <= 2) {

         return n;

      }

      int a = 1, b = 2;

      for (int i = 3; i <= n; i++) {

         int c = a + b;

         a = b;

         b = c;
```

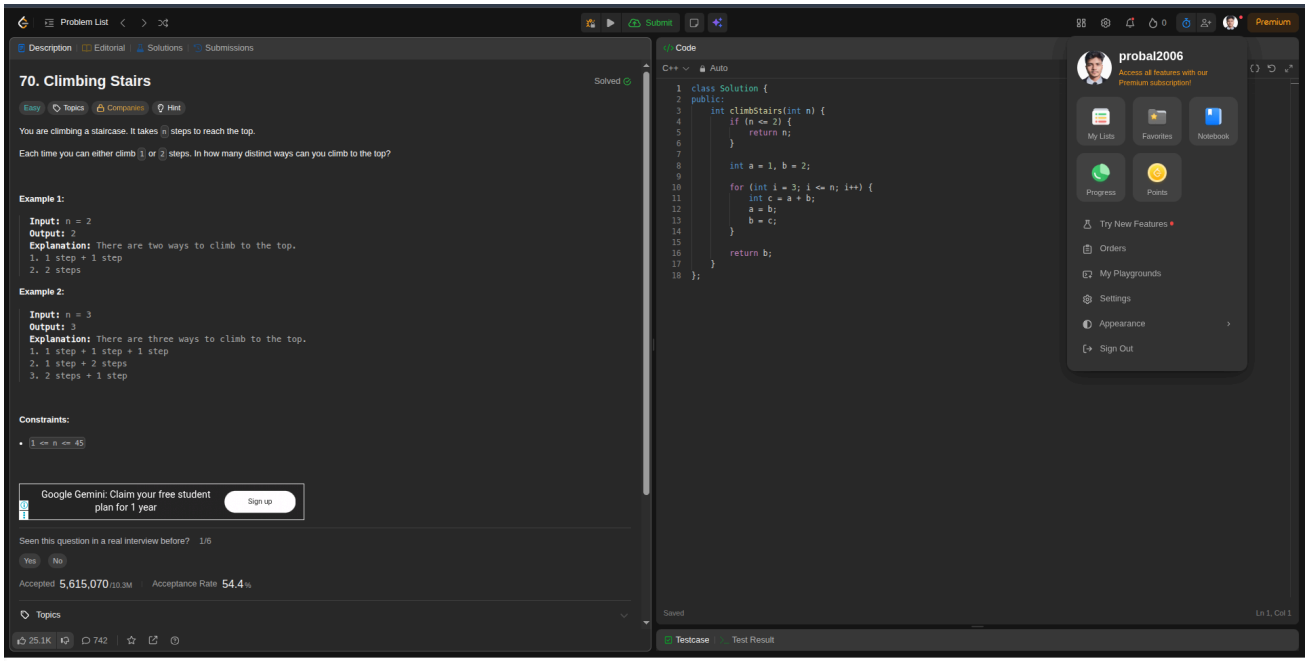
```
        }

        return b;

    }

};
```

Output:



Observations/Outcome: The program successfully calculates the number of distinct ways to climb the staircase using an iterative dynamic programming approach.

Time Complexity: $O(n)$

Space Complexity: $O(1)$

Learning outcomes :

- Understand the basic concept of dynamic programming.
- Apply the Fibonacci sequence to solve a problem.
- Implement an iterative solution in C++.
- Analyze time and space complexity.

Experiment No: 8.2

Aim: To implement a C++ program to find the minimum energy required for a frog to reach the last stair.

Objective: To understand dynamic programming and use it to minimize the energy cost of jumping between stairs.

Problem Statement: A frog is initially at the first stair and wants to reach the last stair. The frog can jump up to K stairs at a time. The energy required for a jump from stair i to stair j is abs(height[i] - height[j]). Find the minimum energy required to reach the last stair.

LeetCode Link: https://www.naukri.com/code360/problems/frog-jump_3621012?utm_source=chatgpt.com

Procedure/Algorithm/Pseudocode:

- Initialize dp[0] = 0.
- Iterate through each stair from 1 to n-1.
- For each stair, consider jumps from 1 to k steps backward.
- Calculate the energy required for each valid jump.
- Store the minimum energy in dp[i].
- Return dp[n-1].

Code:

```
#include <bits/stdc++.h>

using namespace std;

int frogJump(int n, vector<int> &height) {

    // dp[i] will store the minimum energy to reach the i-th stair

    vector<int> dp(n, INT_MAX);

    // Base case: 0 energy needed to start at the first stair

    dp[0] = 0;

    for (int i = 1; i < n; i++) {

        // Jump 1 step forward

        int jumpOne = dp[i - 1] + abs(height[i] - height[i - 1]);

        // Jump 2 steps forward (only if valid)

        int jumpTwo = INT_MAX;

        if (i - 2 >= 0) {

            jumpTwo = dp[i - 2] + abs(height[i] - height[i - 2]);

        }

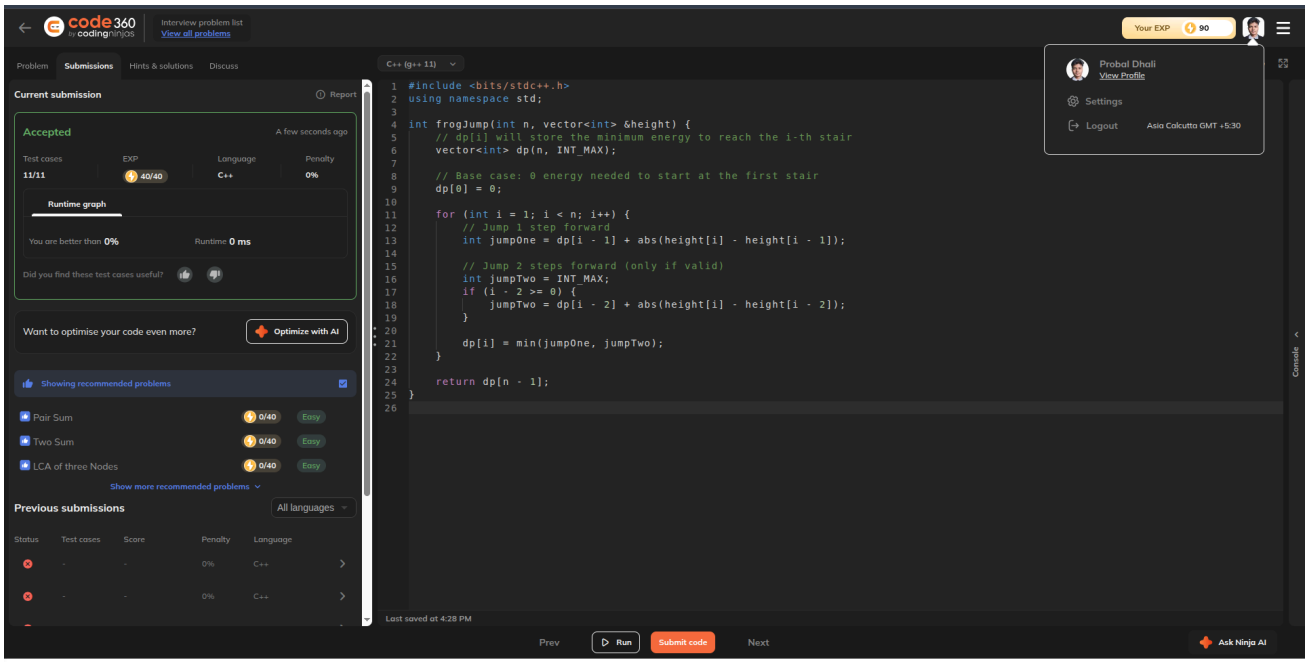
        dp[i] = min(jumpOne, jumpTwo);
```

```
    }

    return dp[n - 1];

}
```

Output:



Observations/Outcome: The program successfully calculates the minimum energy required to reach the last stair using dynamic programming.

Time Complexity: $O(nk)$

Space Complexity: $O(n)$

Learning outcomes :

- Understand dynamic programming for optimization problems.
- Calculate the minimum cost using multiple possible jumps.
- Implement nested loops and the min() function in C++.
- Analyze time and space complexity.